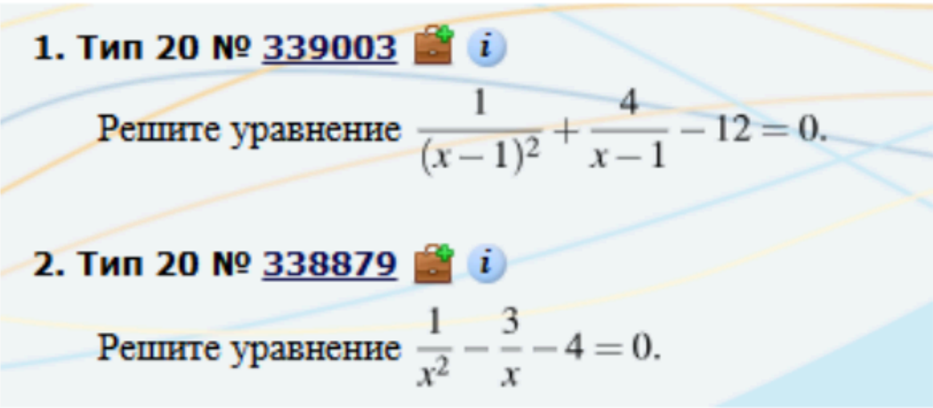
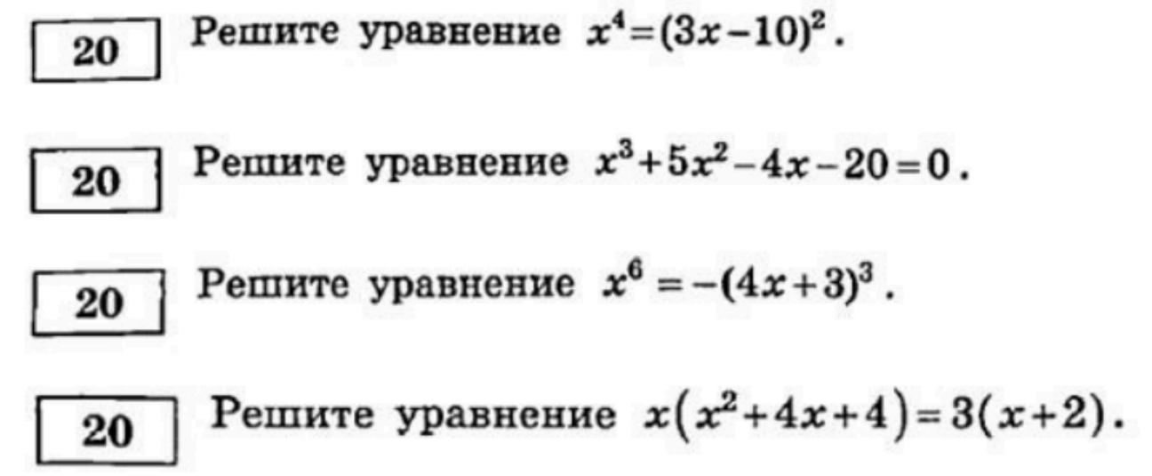


1. Тип 20 № 339003

Решите уравнение $\frac{1}{(x-1)^2} + \frac{4}{x-1} - 12 = 0.$

2. Тип 20 № 338879

Решите уравнение $\frac{1}{x^2} - \frac{3}{x} - 4 = 0.$



20 Решите уравнение $x^4 = (3x-10)^2$.

20 Решите уравнение $x^3 + 5x^2 - 4x - 20 = 0$.

20 Решите уравнение $x^6 = -(4x+3)^3$.

20 Решите уравнение $x(x^2+4x+4) = 3(x+2)$.

Ответы:

$$\frac{1^{\backslash 1}}{(x-1)^2} + \frac{4^{\backslash (x-1)}}{(x-1)} = 12$$

$$\frac{1 + 4x - 4}{(x-1)^2} = \frac{12}{1}$$

$$1 + 4x - 4 = 12(x-1)^2$$

$$1 + 4x - 4 = 12(x^2 - 2x + 1)$$

$$\underbrace{1 + 4x - 4} \underset{\rightarrow}{=} 12x^2 - 24x + 12$$

$$12x^2 - 28x + 15 = 0$$

$$D = 784 - 4 \cdot 12 \cdot 15 = 784 - 720 = 64$$

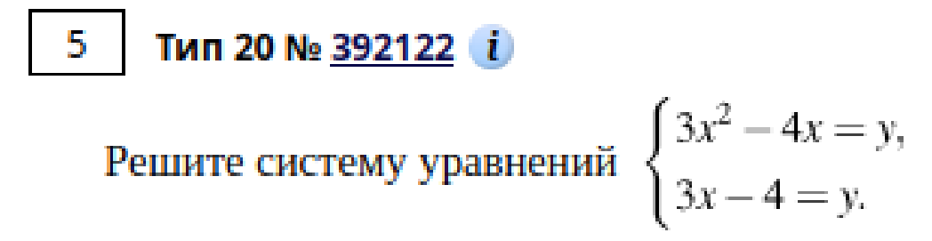
$$x_1 = \frac{28+8}{24} = 1{,}5$$

$$x_2 = \frac{28-8}{24} = \frac{20}{24} = \frac{5}{6}$$

Проверка: на 0 делить нельзя $\Rightarrow x \neq 1$

$x_1 \neq 1;\ x_2 \neq 1$

Ответ: $1{,}5;\ \frac{5}{6}$



5 **Тип 20 № 392122**

Решите систему уравнений $\begin{cases} 3x^2 - 4x = y, \\ 3x - 4 = y. \end{cases}$

$$3x^2 - 4x = 3x - 4$$

$$3x^2 - 7x + 4 = 0$$

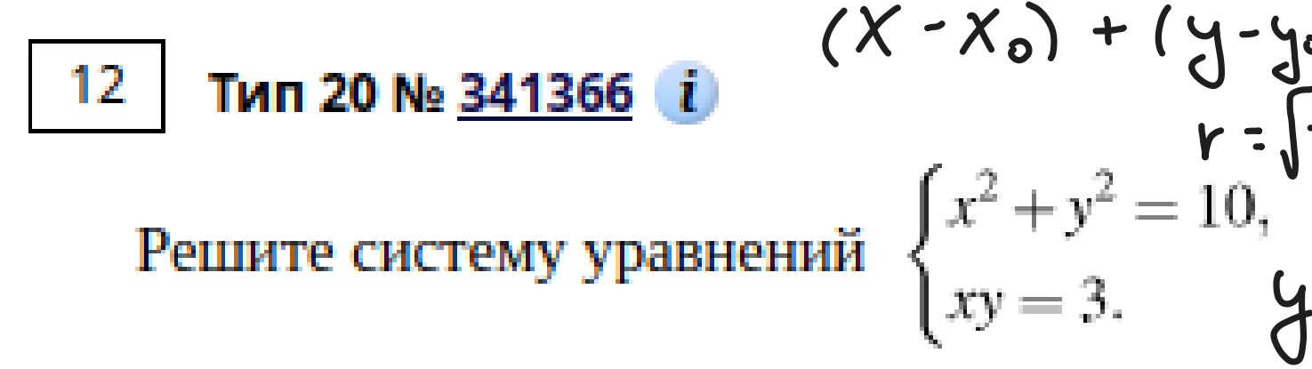
$$D = 49 - 4 \cdot 3 \cdot 4 = 49 - 48 = 1$$

$$x_1 = \frac{7+1}{6} = \frac{4}{3} = 1\frac{1}{3} \Rightarrow y_1 = 0$$

$$x_2 = \frac{7-1}{6} = 1 \Rightarrow y_2 = -1$$

Ответ: $(1\frac{1}{3};0)$

$(1;-1)$



12 **Тип 20 № 341366**

Решите систему уравнений $\begin{cases} x^2 + y^2 = 10, \\ xy = 3. \end{cases}$

$$\begin{cases} x^2 + y^2 = 10 \\ xy = 3 \end{cases} \Rightarrow x = \frac{3}{y}$$

$$x^2 + 2xy + y^2$$

$$\begin{cases} x^2 + y^2 = 10 \\ xy = 3 \quad / \cdot 2 \end{cases}$$

$$+\begin{cases} x^2 + y^2 = 10 \\ 2xy = 6 \end{cases}$$

$$x^2 + 2xy + y^2 = 16$$

$$(x+y)^2 = 16$$

$x + y = 4$ $\qquad$ $x + y = -4$

$y = 4 - x$ $\qquad$ $y = -4 - x$

$$(x - \overset{0}{\overset{\shortparallel}{x_0}})^2 + (y - \overset{0}{\overset{\shortparallel}{y_0}})^2 = r^2$$

$$r = \sqrt{3^2 + 1^2}$$

$$y = \frac{3}{x}$$

$$\left(\frac{3}{y}\right)^2 + y^2 = 10$$

$$\frac{9^{\backslash 1}}{y^2} + y^{2\backslash y^2} = 10$$

$$\frac{9 + y^4}{y^2} = 10$$

$$9 + y^4 = 10y^2$$

$$y^4 - 10\boxed{y^2} + 9 = 0$$

биквадратное уравнение

Замена: $y^2 = t$

$$t^2 - 10t + 9 = 0$$

$$D = 100 - 36 = 64$$

$$t_1 = \frac{10+8}{2} = 9$$

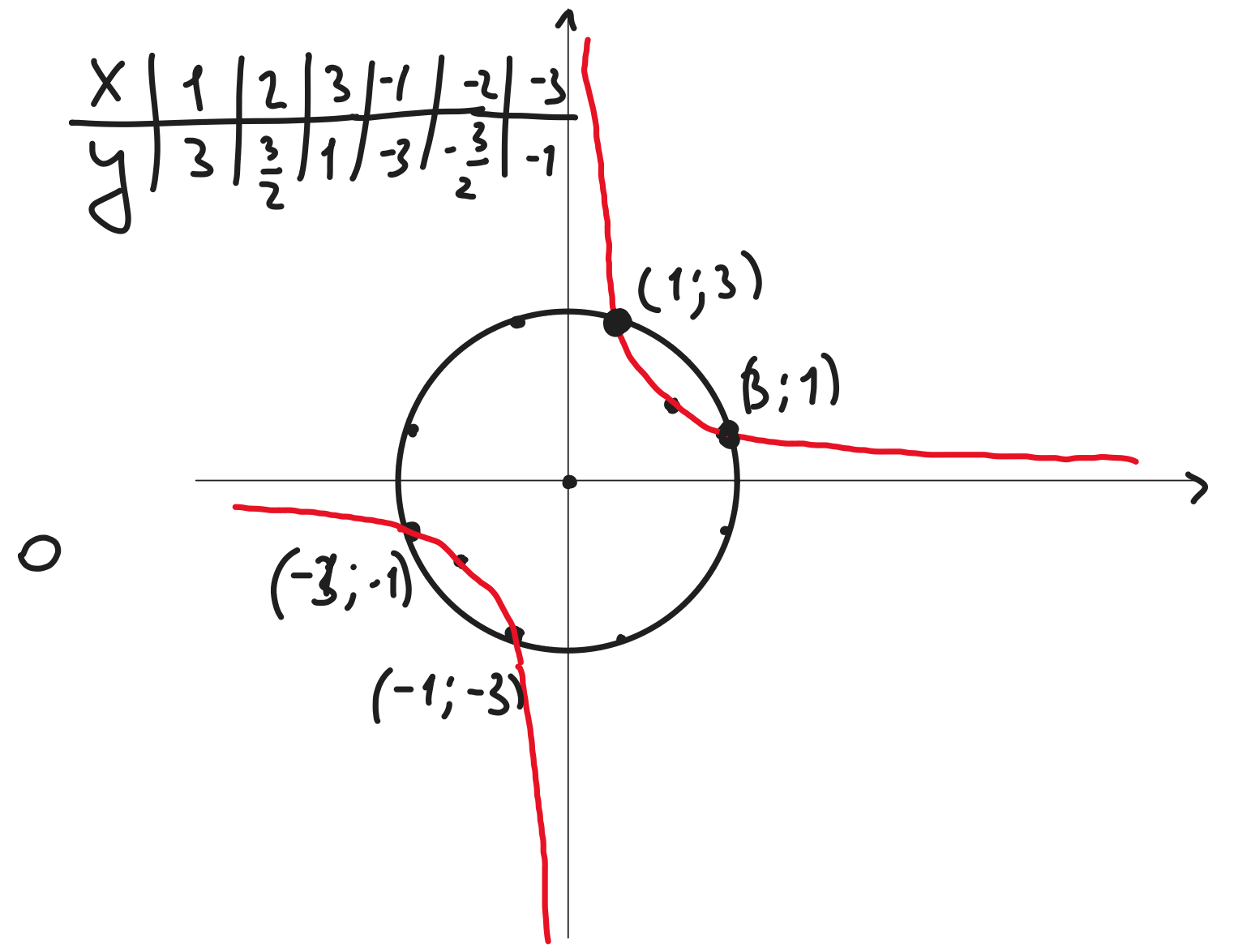
$$t_2 = \frac{10-8}{2} = 1$$

Обратная замена:

$y^2 = 9$ $\qquad$ $y^2 = 1$

$y_1 = 3$ $\qquad$ $y_3 = 1$

$y_2 = -3$ $\qquad$ $y_4 = -1$



x	1	2	3	-1	-2	-3
y	3	$\frac{3}{2}$	1	-3	$-\frac{3}{2}$	-1

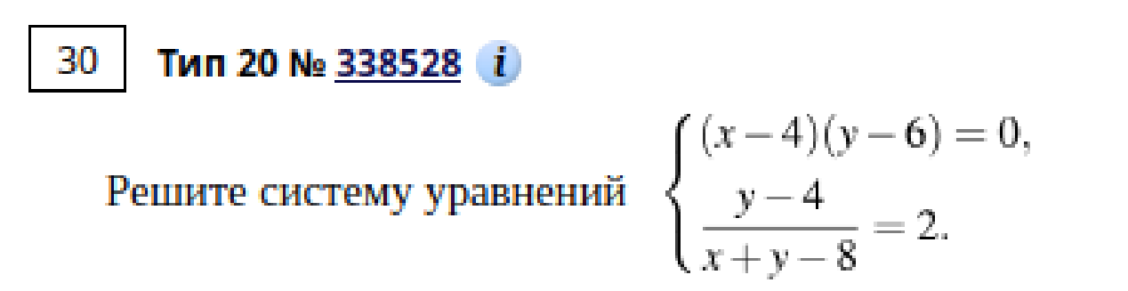
$x = \frac{3}{y}$: $\quad x_1 = \frac{3}{3} = 1$

$x_2 = \frac{3}{-3} = -1$

$x_3 = \frac{3}{1} = 3$

$x_4 = \frac{3}{-1} = -3$

Ответ: $(1;3)$, $(-1;-3)$, $(3;1)$, $(-3;-1)$



30 **Тип 20 № 338528**

Решите систему уравнений $\begin{cases} (x-4)(y-6) = 0, \\ \frac{y-4}{x+y-8} = 2. \end{cases}$

$x - 4 = 0$ $\qquad$ $x = 4$

$y - 6 = 0$ $\qquad$ $y = 6$

$x + y - 8 \neq 0$

$y \neq -x + 8$

$$y - 4 = 2(x + y - 8)$$

$$y - 4 = 2x + 2y - 16$$

$$-y - 2x + 12 = 0$$

$$y = -2x + 12$$

$y = kx\boxed{+b}$

$$(x-4)(-2x + 12 - 6) = 0$$

$x - 4 = 0$ или $-2x + 6 = 0$

$x_1 = 4$ $\qquad$ $x_2 = 3$

$y_1 = 4$ $\qquad$ $y_2 = 6$

Проверха: на 0 делить нельзя $\Rightarrow x + y - 8 \neq 0$

$x_1 = 4,\ y_1 = 4$ $\qquad$ $4 + 4 - 8 = 0 \Rightarrow (4;4)$ – не решение

$x_2 = 3;\ y_2 = 6$ $\qquad$ $3 + 6 - 8 = 1 \neq 0 \Rightarrow (3;6)$ – решение

Ответ: $(3;6)$